Latent Variable Moderated Mediation Using Correct and Misspecified Models

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Abstract

Moderated mediation is seldom estimated with latent variables, and approaches that can estimate it are rarely compared under misspecification. In a preregistered simulation we evaluated three approaches under a latent moderated mediation model, local structural-after-measurement (LSAM) and two system-wide methods, across sample size, interaction magnitude, reliability, predictor distribution, and nine (mis)specification conditions. Under correct specification with normal predictors, all recovered the interaction coefficient and index of moderated mediation, differing modestly in bias, root mean square error, coverage, standard errors, Type I error, and power. Under misspecification, LSAM was most robust, leaving some estimates unaffected by omissions that biased system-wide approaches, although most misspecifications degraded performance.

Keywords: moderated mediation, structural equation modeling, latent interaction, measurement error, model misspecification

Latent Variable Moderated Mediation Using Correct and Misspecified Models

Mediation analysis is a well-known procedure in the behavioral and social sciences (Charlton et al., 2021; Hayes, 2018; Pieters, 2017; Preacher et al., 2007). In its basic and standard form, the effect of an independent variable on an outcome is decomposed into a direct effect and an indirect effect transmitted through a mediator, with all variables observed. This single mediator model has since been extended in a number of directions, most relevant here being the case in which the indirect effect is allowed to depend on a moderator (Edwards & Lambert, 2007; Hayes, 2015; MacKinnon et al., 2007; Montoya, 2024; Preacher et al., 2007).

In this context, a fundamental concern may arise from measurement. Many constructs under investigation are not directly observed but are measured indirectly through multiple items, that is, they are latent variables (Bollen & Hoyle, 2012). When such constructs are reduced to summary scores (e.g., a sum or mean of the items) and analyzed as though they were observed, a restrictive measurement structure is imposed rather than tested (McNeish & Wolf, 2020) and measurement error is left unmodeled. This error can attenuate or inflate the indirect effect and distort inferences about it (Cole & Preacher, 2014; Hoyle & Kenny, 1999), and it is especially consequential for the mediator, which acts at once as a predictor and an outcome (Gonzalez & MacKinnon, 2021). The problem is compounded under moderation, where the product term is even less reliable than either of its constituents (Bohrnstedt & Marwell, 1978; Busemeyer & Jones, 1983). To address this, structural equation modeling (SEM) with latent variables has been advocated, especially in the context of mediation (Cheung & Lau, 2008; Falk & Biesanz, 2015; Ledgerwood & Shrout, 2011; Montoya, 2024). It estimates structural relationships between common factors, from which indicator-specific unique variance has been partitioned, provided the measurement model is correctly specified (Bollen, 1989; DeShon, 1998; Rhemtulla et al., 2020). Yet we also note that latent variables do not necessarily improve power in general, whether for interactions or not (Cham et al., 2012; Ledgerwood & Shrout, 2011; Wang & Rhemtulla, 2021).

Moderated mediation with latent variables is not straightforward in terms of statistical

estimation. Indeed, a latent interaction[1] involves the product of two latent variables, which is unobserved and nonnormal even when its constituents are normal (Kelava & Brandt, 2022). Following this, a number of methods have been proposed to estimate nonlinear effects among latent variables (e.g., Boker et al., 2023; Bollen, 1995; Kenny & Judd, 1984; Klein & Moosbrugger, 2000; Klein & Muthén, 2007; Marsh et al., 2004; Mooijaart & Bentler, 2010). Most adopt a joint ("system-wide") estimation strategy, in which the measurement and structural parameters are estimated simultaneously. This is the case of the two best-known methods, latent moderated structural equations (LMS) and the unconstrained product indicator (UPI) approach. Opposed to these are structural-after-measurement (SAM) approaches (Cox & Kelcey, 2021; Holst & Budtz-Jørgensen, 2020; Ng & Chan, 2020; Rosseel et al., 2025; Wall & Amemiya, 2000, 2003), in which the measurement model is estimated first and the structural model afterward with the measurement parameters held fixed. Many of these approaches have so far rarely been applied in practice, likely held back by limited software, slow estimation, and error-prone implementations. With recent open-source implementations (Holst & Budtz-Jørgensen, 2020; Rosseel et al., 2025; Slupphaug et al., 2025) and non-commercial software (Keller, 2025), this is likely to change.

Beyond estimation, a further consideration concerns interpretation. Mediation is a hypothesis about causal effects, and causal interpretation requires identification assumptions that *may* not be verified from the data alone (Imai et al., 2010; Morelli et al., 2026; Rohrer et al., 2022). With latent variable models, further caution is needed (Sterner et al., 2024; VanderWeele & Vansteelandt, 2022). We focus on one that is central to interpretation: the estimated latent variables must validly represent the intended constructs (Gonzalez & MacKinnon, 2021). In fact, when they do not, the model is subject to interpretational confounding (Burt, 1976), a discrepancy between the empirical meaning assigned to a latent variable through its estimated measurement parameters and the meaning intended by the researcher. This reflects an underlying property of

[1] We use interaction and moderation interchangeably, as the focus is on estimation, but we highlight that they imply different interpretations and causal assumptions (VanderWeele, 2009). An interaction also need not be a product term, although that is the most common functional form used by researchers (Rimpler et al., 2026).

system-wide estimation, namely that the consequences of a misspecification are not confined to the part of the model in which it occurs (Bainter & Bollen, 2014). Burt (1976) proposed estimating the measurement model first, as a SAM approach does (e.g., Rosseel & Loh, 2024), so that the estimated latent variables and their meaning do not depend on the structural model that follows, although this does not imply that the structure is correct (VanderWeele & Vansteelandt, 2022).

To illustrate interpretational confounding, we specify a population mediation model with three latent variables, X, M, and Y, each with three indicators and all loadings equal to 1, and derive its model-implied covariance matrix. We then set the covariances of the M indicators with x_1, x_2, and x_3 to 3, 1, and 1, rather than the implied value of 2, so their association with M no longer follows a single common factor. We fit the model to this matrix ($N = 200$) with and without the mediator, by system-wide SEM and by SAM, and compare the estimated loadings of x_2 and x_3. Without the mediator, both approaches recover the true loadings of 1. Once the mediator enters, the system-wide loadings drop to 0.41, whereas the SAM loadings, estimated before the structural model, remain at 1. The code for this illustration is included in the manuscript source in the project repository.

Despite this, prior simulation work has examined latent interaction estimation mainly under distributional misspecification (Brandt et al., 2014; e.g., Cham et al., 2012; Lonati et al., 2025; Vieira et al., 2026), and in only two cases under misspecification at the measurement level (Brandt et al., 2020; Navarro & Schermelleh-Engel, 2025), which between them considered only the UPI and LMS among the approaches studied here in limited settings. Moreover, when it comes to (moderated) mediation, despite its popularity in applied research, only a few studies have investigated it with latent variables, and those assumed a correctly specified (and saturated) model (Cheung & Lau, 2017; Falk & Biesanz, 2015; Feng et al., 2020). In this article, we therefore evaluate three estimation approaches for a latent variable moderated mediation model: local SAM (LSAM; Rosseel et al., 2025), LMS (Kelava et al., 2011; Klein & Moosbrugger, 2000), and UPI (Kelava & Brandt, 2009; Marsh et al., 2004). These approaches have been explained extensively in previous research and we refer readers to it (e.g., Brandt et al., 2020; Kelava et al., 2011; Lin et

al., 2010; Lonati et al., 2025; Rosseel et al., 2025; Slupphaug et al., 2025; Vieira et al., 2026).

Present Research and Contributions

This study makes two contributions. First, we evaluate LSAM alongside LMS and UPI under both measurement and structural misspecification, in a considerably broader design than previous studies, detailed in the next section. The robustness of LSAM to model misspecification has itself been examined only in linear models (Rosseel & Loh, 2024), with later research disputing it for measurement misspecifications and expecting it to hold mainly for structural misspecifications in non-saturated structural models (Robitzsch, 2022), a case not yet tested for models with nonlinear effects. Second, we obtain inference for the index of moderated mediation (imm; Hayes, 2015), a nonlinear function of the structural parameters, with the Monte Carlo method (MacKinnon et al., 2004; Preacher & Selig, 2012) rather than the bootstrap used in the latent variable studies above (Cheung & Lau, 2017; Feng et al., 2020). Bootstrapping is computationally expensive, and its variants differ modestly in performance and rank differently depending on the criterion (Falk & Biesanz, 2015; Tibbe & Montoya, 2022), whereas the Monte Carlo method needs only the estimates and their covariance matrix. Whether it performs adequately for the imm in the context under investigation, which has not been examined, is a question we address here.

Simulation Design and Performance Evaluation

In this section we describe the design of the simulation study and the measures used to evaluate performance, following the preregistration and some recent recommendations for the conduct and reporting of simulation studies (Pawel et al., 2026; Siepe et al., 2024).

Population Model and Estimands

We study a moderated mediation model among four latent variables: an exogenous predictor X, a moderator W, a mediator M, and an outcome Y. The structural model is as follows

$$M = a_1 X + a_2 W + a_3 \, (X \times W) + \zeta_M, \qquad Y = b \, M + c' \, X + \zeta_Y.$$

The primary estimands are the interaction coefficient (a_3) and the imm (i.e., a_3b). The remaining structural coefficients (a_1, a_2, b, c') and the twelve freely estimated factor loadings serve as secondary estimands. Note the imm is a derived quantity rather than a free parameter, hence its true value is computed as a_3b implied by the correctly specified model.

Population values were fixed at $a_1 = 0.4$, $a_2 = 0.3$, $b = 0.5$, and $c' = 0.15$, with $\mathrm{Cor}(X, W) = 0.3$. The interaction coefficient a_3 is a design factor (see below). These coefficients, together with the structural disturbance variances, were chosen so that under the baseline condition (normal exogenous predictors and a correctly specified model) each dependent latent variable has $R^2 \approx 0.30$ at the interaction level $a_3 = 0.20$. Also note that, because a_3 contributes to the variance of M, the explained variance is naturally higher with a larger a_3 and lower when its effect is null. Each latent variable is measured by four congeneric indicators with loadings 1.0, 0.8, 0.7, and 0.6. Figure 1 shows the population model, where the measurement and structural misspecifications studied below are drawn in gray.

Simulation Conditions

We employed a fully factorial design crossing five factors: sample size ($N \in \{100, 200, 300, 500\}$), interaction magnitude ($a_3 \in \{0, 0.20, 0.40\}$), measurement reliability (low = 0.5, high = 0.7), the distribution of the exogenous latent predictors (normal, uniform, t_5, and two χ^2 conditions with same- or opposite-direction skewness), and the (mis)specification conditions (nine cases; see Data Generation below). This yielded $4 \times 3 \times 2 \times 5 \times 9 = 1{,}080$ conditions, each replicated 1,000 times.

Sample sizes cover the small to moderate range, where interaction terms are estimated with limited power (Brandt et al., 2020; Lonati et al., 2025). The interaction magnitudes include a null condition to assess Type I error for the interaction coefficient and imm, and two nonzero levels representing small and moderate latent moderation effects, following previous research (Feng et al., 2020; Klein & Moosbrugger, 2000). Reliability levels of 0.5 and 0.7 were chosen because measurement error is amplified in product terms and is therefore especially consequential for latent interaction estimation (Brandt et al., 2020; Lonati et al., 2025; Vieira et al., 2026).

Nonnormality of the latent predictors is both empirically common and methodologically central for latent interaction approaches, whose distributional assumptions differ across approaches (Cham et al., 2012; Klein & Moosbrugger, 2000; Klein & Muthén, 2007; Lonati et al., 2025; Vieira et al., 2026). The misspecification conditions follow the moderated mediation framework for the two structural omissions (Preacher et al., 2007) and prior latent interaction simulation work for the two measurement omissions (Brandt et al., 2020). They were also chosen as interpretable violations, acting respectively as a measured mediator–outcome confounder (Imai et al., 2010), a conflated moderated direct effect (Edwards & Lambert, 2007), exposure-correlated measurement error in the mediator (Fritz et al., 2016; VanderWeele et al., 2012), and a shared unmeasured cause of a mediator and an outcome indicator (Cole et al., 2007). The misspecification cases are detailed below.

Estimation Approaches

We used LSAM (Rosseel et al., 2025; Rosseel & Loh, 2024; Vieira et al., 2026), with the measurement model estimated by maximum likelihood (ML) in the first step and with standard errors obtained as in Vieira et al. (2026) (see their Appendix A). In this case, the interacting predictors X and W shared a single measurement block, while M and Y formed their own blocks. We further used LMS (Klein & Moosbrugger, 2000), with robust standard errors, and the extended UPI approach with double mean centering (Kelava & Brandt, 2009; Lin et al., 2010; Marsh et al., 2004), estimated with robust ML, that is, with Huber–White sandwich standard errors.

Data Generation

Appendix describes the complete data-generating mechanism in detail. Briefly, the latent variables X and W were drawn from a bivariate distribution with $\mathrm{Cor}(X, W) = \rho = 0.3$, constructed with the vine-to-anything (VITA) algorithm (Grønneberg et al., 2022). Five exogenous distributions were examined, each standardized to mean 0 and variance 1. Thus, the conditions differ only in distributional shape, not in scale or location. Specifically, the normal condition draws $X, W \sim \mathcal{N}(0, 1)$ through a forced Gaussian copula, whereas the remaining four combine nonnormal margins with the Clayton copula: uniform, t_5, and two $\chi^2(2)$ conditions in

which X and W are skewed in the same or in opposite directions. Then, M and Y were generated from the structural equations above with independent Gaussian disturbances $\zeta_M \sim \mathcal{N}(0, \psi_M)$ and $\zeta_Y \sim \mathcal{N}(0, \psi_Y)$, and the four indicators ($j = 1, \ldots, 4$) of each latent variable $\eta \in \{X, W, M, Y\}$ were formed as $z_{\eta j} = \lambda_j \eta + \delta_{\eta j}$, with loadings $\lambda = (1, 0.8, 0.7, 0.6)$ and Gaussian errors $\delta_{\eta j} \sim \mathcal{N}(0, \theta_{\eta j})$ whose variances $\theta_{\eta j}$ were scaled to the target reliability $\omega \in \{0.5, 0.7\}$.

The nine (mis)specification conditions comprise one correctly specified condition and four types of misspecification, each at two empirically calibrated magnitudes (standardized 0.25 and 0.50, detailed in Appendix), similar to Brandt et al. (2020). Two are structural: an omitted direct effect of the moderator on the outcome (c_2) and an omitted moderated direct effect (c_3). The others are in the measurement model: a cross-loading of a mediator indicator on X (`cl`) and a residual covariance between a mediator and an outcome indicator (`rc`). In every case the effect is present in the population but omitted from the estimated model.

Convergence, Outliers, and Inadmissible Solutions

Following previous research (Lonati et al., 2025; Vieira et al., 2026), replications were screened before analysis in two sequential steps. In both, a replication that failed the screening for any one estimator was removed for all three, so that every estimator is compared on an identical set of replications. First, we checked convergence. A solution was treated as nonconverged when any estimate, standard error, or confidence-interval bound of any structural coefficient or loading was missing or non-finite, and a replication was removed from all approaches whenever any one of them did not converge. Second, among the remaining replications, we screened for outliers, identified per condition, method, and parameter of interest as point estimates more than three interquartile ranges below the first or above the third quartile, together with replications whose estimated standard error exceeded 10 on any parameter of interest[2]. Again, a replication identified as an outlier for any estimation approach was removed from all of them. We also identified

[2] We added the standard error criterion because some replications retained anomalous standard errors despite acceptable point estimates. It additionally removed 125 replications (about 0.01% of the globally converged ones), predominantly for UPI at the smallest sample size with low reliability and skewed χ^2 predictors.

inadmissible solutions, where any residual (unique) variance was nonpositive or the covariance matrix of the exogenous latent variables and latent disturbances was not positive definite.

Performance Measures

Performance was assessed with the measures implemented in the *simhelpers* package (Joshi & Pustejovsky, 2025), computed for the imm, the structural coefficients, and the loadings. These measures, with their empirical and Monte Carlo standard error (MCSE) calculations, are listed in Table 1. Note that the MCSEs are obtained in closed form where available and by jackknife otherwise. In addition, where applicable, we report the rejection rate (Type I error when $a_3 = 0$, power otherwise).

Inference for the imm followed the Monte Carlo method as mentioned before: for $d = 1, \ldots, D$ (with $D = 20{,}000$) we drew

$$\left(a_3^{(d)}, b^{(d)}\right) \sim \mathcal{N}_2\left((\hat{a}_3, \hat{b}),\, \hat{\Sigma}_{a_3 b}\right), \qquad \text{imm}^{(d)} = a_3^{(d)}\, b^{(d)},$$

where $\hat{\Sigma}_{a_3 b}$ is the estimated covariance of $(\hat{a}_3, \hat{b})$, and took the 2.5th and 97.5th empirical percentiles of $\{\text{imm}^{(d)}\}_{d=1}^{D}$ as the interval. The structural coefficients and loadings use Wald-based intervals from the same estimated covariance matrix of the parameter estimates.

Reproducibility and Computational Details

All analyses were conducted in R (R Core Team, 2025) using the *lavaan* (Rosseel, 2012) and *modsem* (Slupphaug et al., 2025) packages for estimation, *covsim* (Grønneberg et al., 2022) and *rvinecopulib* (Nagler & Vatter, 2025) for data generation, and *simhelpers* (Joshi & Pustejovsky, 2025) for performance evaluation. Conditions were run in parallel with the *parallel* package. Reproducible and independent random-number streams were obtained with the L'Ecuyer-CMRG generator. The computational environment was pinned with Nix (Dolstra et al., 2004) via the *rix* package (Rodrigues & Baumann, 2026) to ensure reproducibility, and the article was written with the *apaquarto* extension (Schneider, 2026). For the R version, package versions, code, manuscript, and materials, see the project repository.

Results

We present the results in three sections. First, the results for the interaction coefficient and the imm are reported. We then turn to the remaining structural coefficients (i.e., a_1, a_2, b, and c'), and we close with the estimated factor loadings, focusing mainly on that of the cross-loaded indicator (m_3). Due to the large number of conditions, we focus on a subset. In most figures we fix the interaction coefficient at 0.2 and show $N = 200$ and 500 for the interaction coefficient and imm, and solely $N = 500$ for the structural coefficients and loadings. For the latter two, we further restrict to high reliability, whereas the interaction coefficient and imm are shown at both reliability levels. All results and detailed convergence and outlier information are available in the repository.

Convergence, Outliers, and Inadmissible Solutions

Convergence was high throughout. LSAM converged in every replication across all conditions, and LMS in at least 99.9%. UPI was the least stable, with its convergence rate falling to about 92% in the worst condition (the smallest sample size, low reliability, and skewed χ^2 predictors). At high reliability, however, UPI converged in at least 99% of replications regardless of sample size, distribution, or misspecification. Outliers were likewise rare, staying below 1% in most conditions and reaching up to roughly 11% for UPI and 8% for LSAM, only at $N = 100$ with low reliability and skewed (χ^2) predictors. The results reported below did not exclude inadmissible solutions, which arose almost exclusively for UPI; excluding them leads to negligible changes, confined to the smallest sample size, low reliability, χ^2 conditions.

Main Structural Estimates

Relative Bias

We interpret values whose absolute relative bias exceeds 0.10 (10%) as biased, following previous research (Brandt et al., 2014; Vieira et al., 2026). Under correct specification with normal predictors, all three approaches recover both the interaction coefficient and imm without bias at every sample size and reliability level, and the same holds under uniform and t_5 predictors apart from isolated LMS cells at $N = 100$ with low reliability (Figure 2 and Figure 3). Moreover,

the χ^2 conditions affect all three approaches, but differently. UPI and LSAM are biased only at $N = 100$ with low reliability, where UPI shows the largest deviation, whereas LMS remains biased across all sample sizes at low reliability, negatively under opposite skewness at the larger interaction and positively under same skewness at the smaller one.

Under misspecification, c_2 biases the imm for all three estimation approaches across distributions, and so does c_3 in most conditions, as expected. The size of the bias nevertheless varies with the predictor distribution, being largest under normal and same-skewness χ^2 predictors and smallest under opposite-skewness χ^2 and t_5 predictors. For the interaction coefficient, c_2 produces essentially no bias for all approaches, whereas c_3 leads to bias under the system-wide approaches, mainly at low reliability. Among the measurement omissions, `cl` leaves the imm unbiased except for LMS in the χ^2 conditions, and attenuates the interaction coefficient only under the system-wide methods. Last, `rc` induces an upward bias in the imm at low reliability, most pronounced for LMS under same-skewness χ^2, and leaves the interaction coefficient largely unchanged. For the structural omissions, the bias increases with the omission magnitude wherever present, and it is generally larger at low reliability (Figure 4). We also note that the bias of the interaction coefficient under c_3 is the one result not implied by the omission alone.

Relative RMSE

Under correct specification with normal predictors, the relative RMSE of both the interaction coefficient and the imm falls with sample size and reliability for all three approaches, with LMS the most efficient, LSAM intermediate, and UPI the least (Figure 5). In general, the same holds for both quantities under the nonnormal predictors, except under opposite-skewness χ^2 at low reliability, where LSAM is as inefficient as UPI.

Under misspecification, the structural omissions (c_2 and c_3) inflate the RMSE of the imm for all three approaches. Under c_3 the inflation is clearly larger for the system-wide approaches, especially at low reliability, whereas under c_2 the three are more similar. For the interaction coefficient, c_3 inflates the RMSE for both system-wide approaches, more so at low reliability, while LSAM is unaffected. The measurement omissions (`cl` and `rc`) have a negligible effect,

except for LMS under the medium `cl` at low reliability.

Standard Errors

We interpret SE/SD ratios below 0.90 (underestimation) or above 1.10 (overestimation) as biased (Brandt et al., 2014; Vieira et al., 2026), and similarly for the SE^2/VAR ratio. Note that the standard errors for the imm are based on the Monte Carlo method (Figure 6 and Figure 7). Under correct model specification with normal predictors, the SE/SD ratio of both the interaction coefficient and the imm is unbiased, except for UPI, which overestimates at $N = 200$ with low reliability. Still at $N = 200$ with low reliability, that overestimation extends to the t_5 and χ^2 conditions for UPI and, under opposite-skewness χ^2, to LSAM. Note that with $N = 500$, only UPI under opposite-skewness χ^2 remains overestimated, whereas the interaction coefficient is slightly underestimated for all three approaches under t_5 at high reliability and for LSAM under same-skewness χ^2 at low reliability.

Moreover, in general, model misspecification barely alters this pattern. The same deviations recur throughout all omissions, with some fluctuations. With respect to the SE^2/VAR ratio, a similar pattern is observed across most conditions. Importantly, however, with $N = 200$ at both reliability levels, UPI yielded overestimated values for both χ^2 conditions, while LSAM showed overestimated values for the opposite-skewness χ^2 (figures available in the project repository).

Coverage

Coverage between 92.5% and 97.5% was considered acceptable (Figure 8 and Figure 9). Under correct model specification with normal predictors, coverage is around the nominal level with all three approaches for the interaction coefficient and imm. Still with a correct specification, it slightly drops below the acceptable range under the t_5 condition for all approaches across reliability levels and sample sizes, as well as under same-skewness χ^2 but in this case mainly at $N = 500$ and low reliability.

Under model misspecification, c_2 and c_3 reduce the imm coverage, since its interval is then centered on a biased estimate, with the largest decrease for the system-wide approaches

under c_3 and, under same-skewness χ^2, for LMS under c_2 as well. This is larger with increasing sample size and is present under most distributions, albeit with different magnitudes. Moreover, under t_5 and same-skewness χ^2 predictors, coverage remains at or slightly below the acceptable range under the measurement omissions across sample sizes and reliability levels. For the interaction coefficient, the medium cross-loading (`cl`) is the most damaging case, lowering coverage under the system-wide approaches, while LSAM holds near nominal. We also note that c_3 lowered coverage for the system-wide approaches at $N = 500$, most for LMS at low reliability, whereas c_2 and `rc` leave coverage close to nominal.

Hypothesis Testing (Type I Error and Power)

Type I error rates between 2.5% and 7.5% were considered acceptable (Figure 10). Under correct model specification with normal predictors and $a_3 = 0$, the interaction coefficient and imm hold their nominal level reasonably well for all three approaches, with some negligible conservativeness for UPI at $N = 200$ with low reliability. Under the nonnormal predictors, LMS shows slightly inflated Type I error under t_5 at both reliability levels and under both χ^2 conditions at low reliability, LSAM under t_5 at high reliability, whereas UPI is conservative under opposite-skewness χ^2 and uniform predictors at low reliability.

Under misspecification, this pattern largely persists throughout distributions. LMS shows inflated Type I error in most cells under t_5 and both χ^2 conditions, most severely under the structural omissions and the cross-loading at low reliability, and even under normal predictors for the medium c_3. UPI is instead conservative at low reliability, in several cells at $N = 200$ and throughout under opposite-skewness χ^2, but also inflated under t_5 at high reliability for several omissions, whereas LSAM stays mostly within the acceptable range, with inflation confined largely to t_5 predictors.

At $a_3 = 0.2$ the imm is detected with power that rises with sample size and reliability under normal predictors, with LMS the most powerful (Figure 11). Under the nonnormal predictors, LMS keeps the highest power under χ^2, whereas under t_5 and uniform predictors LSAM matches it, and power falls substantially under opposite-skewness χ^2 predictors, most for

UPI and LSAM. Under misspecification, power barely changes for LSAM, whereas for the system-wide approaches it rises under c_3 and falls under the cross-loading, except for LMS under opposite-skewness χ^2, where its inflated Type I error lifts it. We note, however, that interpreting power in the case of bias and inflated Type I error should be avoided.

Additional Structural Estimates

For the remaining structural coefficients (a_1, a_2, b, and c'), at high reliability and $N = 500$, we report their relative bias (Figure 12), coverage (Figure 13), and SE/SD ratio (Figure 14) across all misspecification conditions and distributions. Under correct specification, all approaches are unbiased with acceptable coverage and SE/SD ratios throughout distributions. Under misspecification, the structural omissions bias the outcome-equation coefficients b and c' for all approaches alike, apart from b under c_3 with opposite-skewness χ^2, whereas the mediator-equation coefficients a_1 and a_2 stay unbiased under LSAM and, for the system-wide approaches, a_2 is biased under c_2. Among the measurement omissions, `cl` is the most disruptive, biasing a_1 and c' for all approaches and additionally a_2 and b for the system-wide approaches, while `rc` leaves all four coefficients within the acceptable range. Where the bias is substantial, it is smaller under LSAM, except for c' under the structural omissions, where the three approaches are similar. Coverage follows the bias, dropping most for b under c_2, for c' under c_3, and for a_1 under `cl`. The SE/SD ratio stays within the acceptable range, except for b and c' under c_3, which are underestimated, most under t_5 predictors for c' and under t_5 and same-skewness χ^2 for b.

Measurement Loadings

For the twelve freely estimated factor loadings, all but one are recovered with acceptable performance across every measure and essentially every condition, the exceptions being noisy deviations of the SE^2/VAR ratio concentrated at $N = 100$ with low reliability. We display only the loading of the cross-loaded mediator indicator m_3, with its relative bias, coverage, SE/SD ratio, and SE^2/VAR ratio at high reliability and $N = 500$ (Figure 15). For m_3, `cl` is the only consequential misspecification. Its relative bias rises sharply under all approaches, less so under LSAM, and its coverage collapses toward zero. The two standard error measures stay within the

acceptable range, except for LSAM under the medium `cl` with t_5 and χ^2 predictors. The remaining omissions leave the loading unaffected.

Discussion

We evaluated three estimation approaches in the context of a latent variable moderated mediation model, the two-step LSAM and the system-wide LMS and UPI, under correct specification and under a large number of structural and measurement misspecifications as well as distributional violations.

Under correct specification with normal predictors, all three recovered the imm and the interaction coefficient without bias and with nominal coverage, differing mainly in the SE/SD ratio, Type I error, relative RMSE, and power. LMS had the lowest relative RMSE and the highest power but, under nonnormal predictors, showed bias under χ^2 predictors at low reliability together with below-nominal coverage and inflated Type I error. LSAM yielded Type I error closest to nominal, at the cost of a higher relative RMSE and lower power in some cases, while UPI fell in between, unbiased except at the smallest sample size with low reliability, where it was also the least stable. Both standard error measures stayed near one except in small samples, most at low reliability, where UPI overestimated standard errors, and UPI and LSAM struggled most under opposite-skewness χ^2 predictors.

Under misspecification, LSAM was the more robust approach. The structural omissions in the outcome equation (c_2, c_3) biased the outcome-side coefficients for all approaches but left the mediator equation, and hence the interaction coefficient, unaffected under LSAM, whereas the system-wide approaches also carried them into the mediator equation, biasing a_2 under c_2 and the interaction coefficient under c_3. The cross-loading (`cl`) was the most damaging throughout. Under the system-wide approaches it biased the interaction coefficient and most structural coefficients and sharply lowered coverage at low reliability, while under LSAM it affected only a_1 and c' and left the coverage of the interaction coefficient near nominal, although its standard errors for the affected loading were underestimated under nonnormal predictors. The residual covariance (`rc`) had little effect. Where bias was substantial, it was smaller under LSAM, except

for c' under the structural omissions, where the three were similar, and it increased with the magnitude of the omission and at lower reliability.

Regarding the imm, the Monte Carlo method delivered standard errors and confidence intervals that behaved like those of the interaction coefficient, with the deviations at low reliability and small samples reflecting the standard errors of the underlying estimates. Under misspecification its coverage dropped with the bias of its constituents despite wider intervals. The method thus appears an adequate and far cheaper alternative to the bootstrap here, although a direct comparison remains to be made.

These patterns are broadly consistent with previous research. The behavior of LMS and UPI under nonnormality and their stability under normality reproduce earlier findings (Brandt et al., 2014; Cham et al., 2012; Lonati et al., 2025; Vieira et al., 2026), as does that of LSAM (Rosseel et al., 2025; Vieira et al., 2026). The measurement misspecification results align with Brandt et al. (2020), where an omitted cross-loading was likewise damaging for every estimator, including UPI. Navarro and Schermelleh-Engel (2025) likewise found the latent interaction effect underestimated under omitted cross-loadings with LMS. Even so, Brandt et al. (2020) found no approach free of bias and none uniformly best, as we also observed. Lastly, the contrast between the structural and the measurement omissions matches the expectation of Robitzsch (2022) that the advantage of SAM lies in structural rather than measurement misspecification, since the cross-loading biased LSAM as well, whereas the structural omissions left its mediator equation untouched.

Finally, although we evaluated the estimation approaches from a statistical perspective, the moderated mediation model itself rests on strong causal assumptions (Imai et al., 2010; Rohrer et al., 2022). Good statistical performance in certain conditions is therefore not, on its own, a justification for a causal claim, and we direct interested readers to the developing literature on the causal interpretation of mediation (Imai et al., 2010; Morelli et al., 2026). Moreover, misspecification had strong consequences across all three approaches, and many of the omissions studied may be highly plausible in applied contexts.

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Table 1

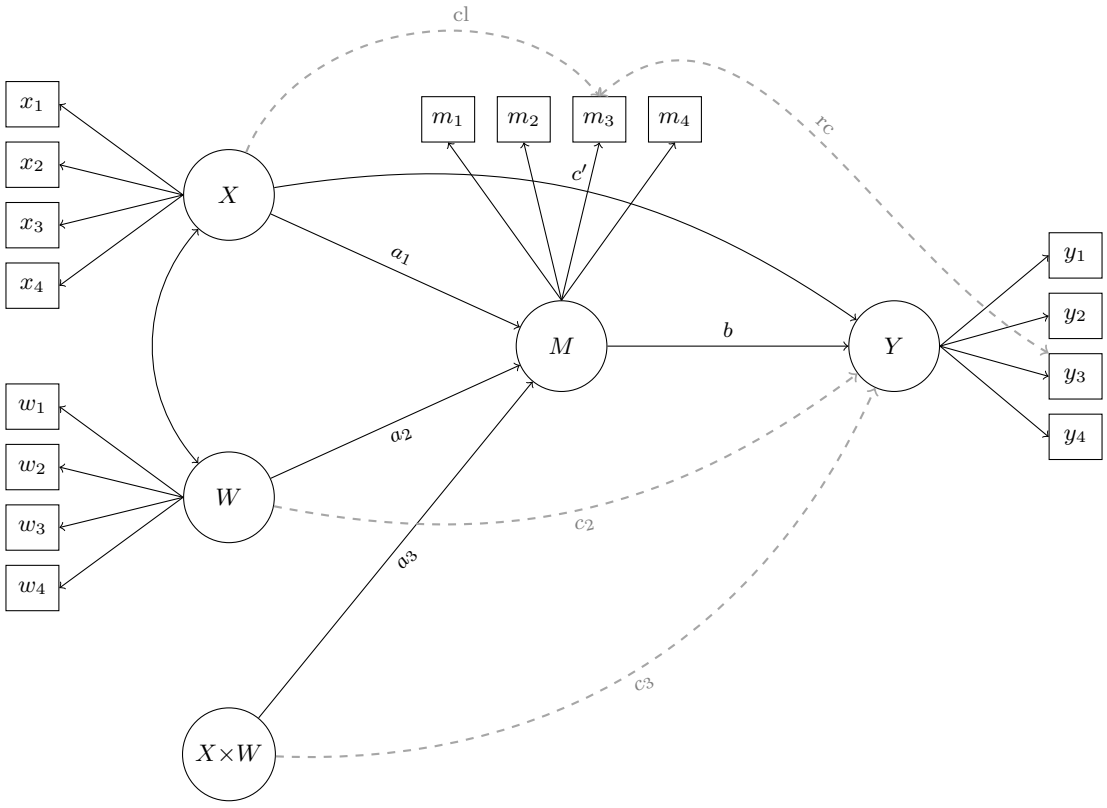
Subset of the Performance Measures for Evaluating Parameter Estimates

Performance measure	Empirical calculation	MCSE
Relative criteria		
Relative bias	$\frac{\bar{\hat{\theta}}}{\theta} - 1$	$\sqrt{S^2_{\hat{\theta}}/(K\theta^2)}$
Relative RMSE	$\frac{1}{\lvert\theta\rvert}\sqrt{\frac{1}{K}\sum_{k=1}^{K}(\hat{\theta}_k - \theta)^2}$	$\frac{1}{\lvert\theta\rvert}\sqrt{\frac{K-1}{K}\sum_{j=1}^{K}(\mathrm{RMSE}_{(j)} - \mathrm{RMSE})^2}$
Standard errors		
Mean SE	$\overline{SE} = \frac{1}{K}\sum_{k=1}^{K}\widehat{SE}_k$	—
SE/SD ratio	$\overline{SE}/S_{\hat{\theta}}$	—
SE^2/VAR ratio	$\nu = \overline{SE^2}/S^2_{\hat{\theta}}$	$\sqrt{\frac{K-1}{K}\sum_{j=1}^{K}(\nu_{(j)} - \nu)^2}$
Coverage		
Coverage	$c = \frac{1}{K}\sum_{k=1}^{K} I(L_k \le \theta \le U_k)$	$\sqrt{c(1-c)/K}$
Hypothesis testing		
Rejection rate	$r = \frac{1}{K}\sum_{k=1}^{K} I(0 \notin [L_k, U_k])$	$\sqrt{r(1-r)/K}$

Note. $\hat{\theta}_k$ is the estimate of parameter θ in replication k ($k = 1, \ldots, K$ usable replications), $\bar{\hat{\theta}}$ its mean, $S^2_{\hat{\theta}}$ its sample variance, and $S_{\hat{\theta}}$ its standard deviation (SD); the root mean square error is $\mathrm{RMSE} = \sqrt{\frac{1}{K}\sum_k(\hat{\theta}_k - \theta)^2}$; $\widehat{SE}_k$ is the estimated standard error (SE), $\overline{SE^2} = \frac{1}{K}\sum_k \widehat{SE}^2_k$, and L_k, U_k the confidence-interval limits; c and r denote coverage and rejection rate, and ν the SE^2/VAR ratio. The subscript (j) marks a leave-one-out jackknife replicate. The MCSEs for relative RMSE and the SE^2/VAR ratio are computed by jackknife; the others are closed-form, and dashes mark MCSEs that are not computed.

Figure 1

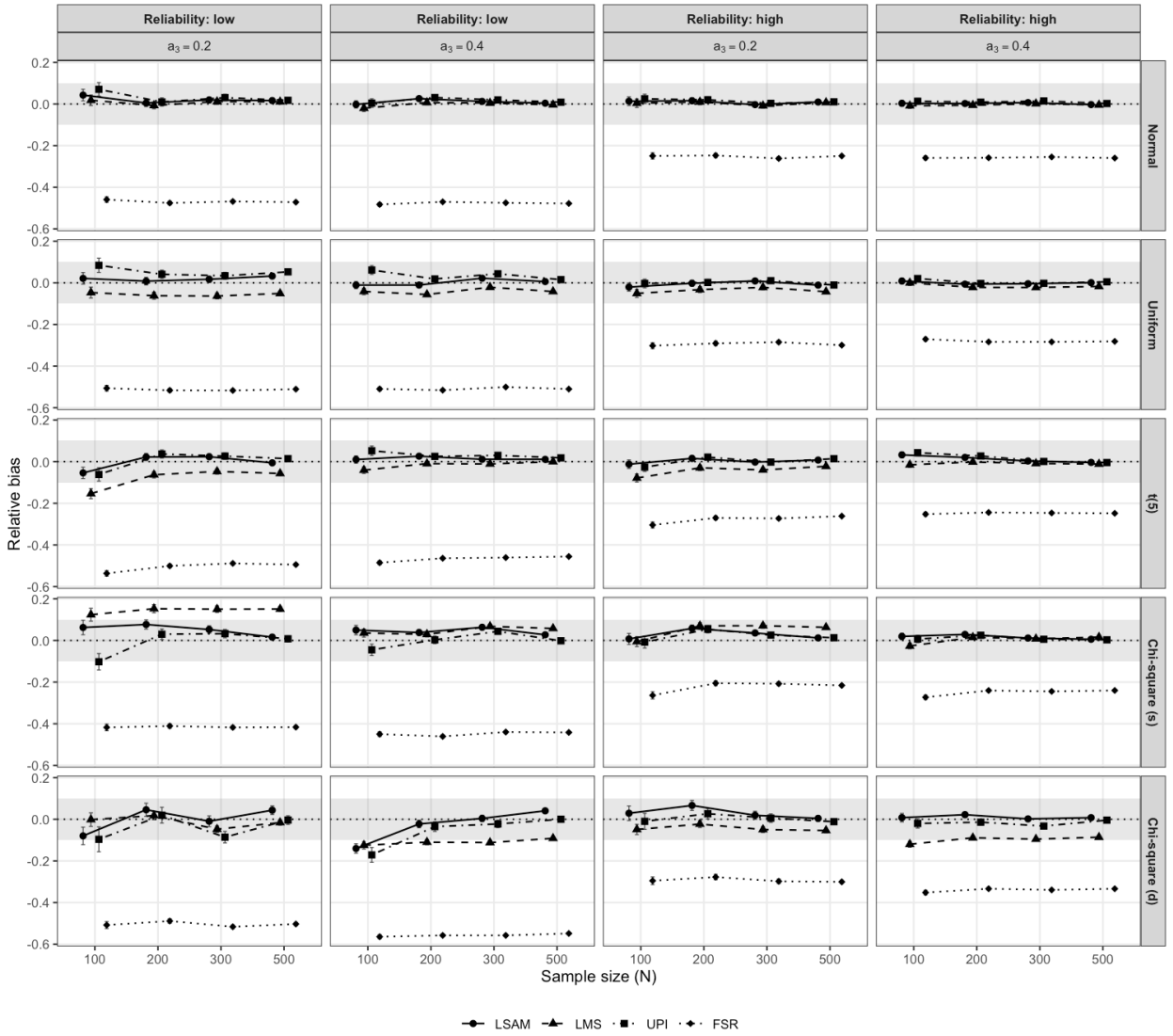
Population Model for the Simulation Study.



Note. X, W, M, and Y are latent variables, each measured by four congeneric indicators. Circles are latent variables and $X{\times}W$ is the latent interaction. Black elements form the correctly specified model that every approach fits (a_1, a_2, a_3, b, c', and the X–W covariance). Gray elements are the misspecifications present in the population but omitted from the fitted model: two in the structural part, an omitted moderator effect on the outcome (c_2) and an omitted moderated direct effect (c_3); and two in the measurement part, an omitted cross-loading of m_3 on X (cl) and an omitted residual covariance between m_3 and y_3 (rc). Residual variances and intercepts are not shown.

Figure 2

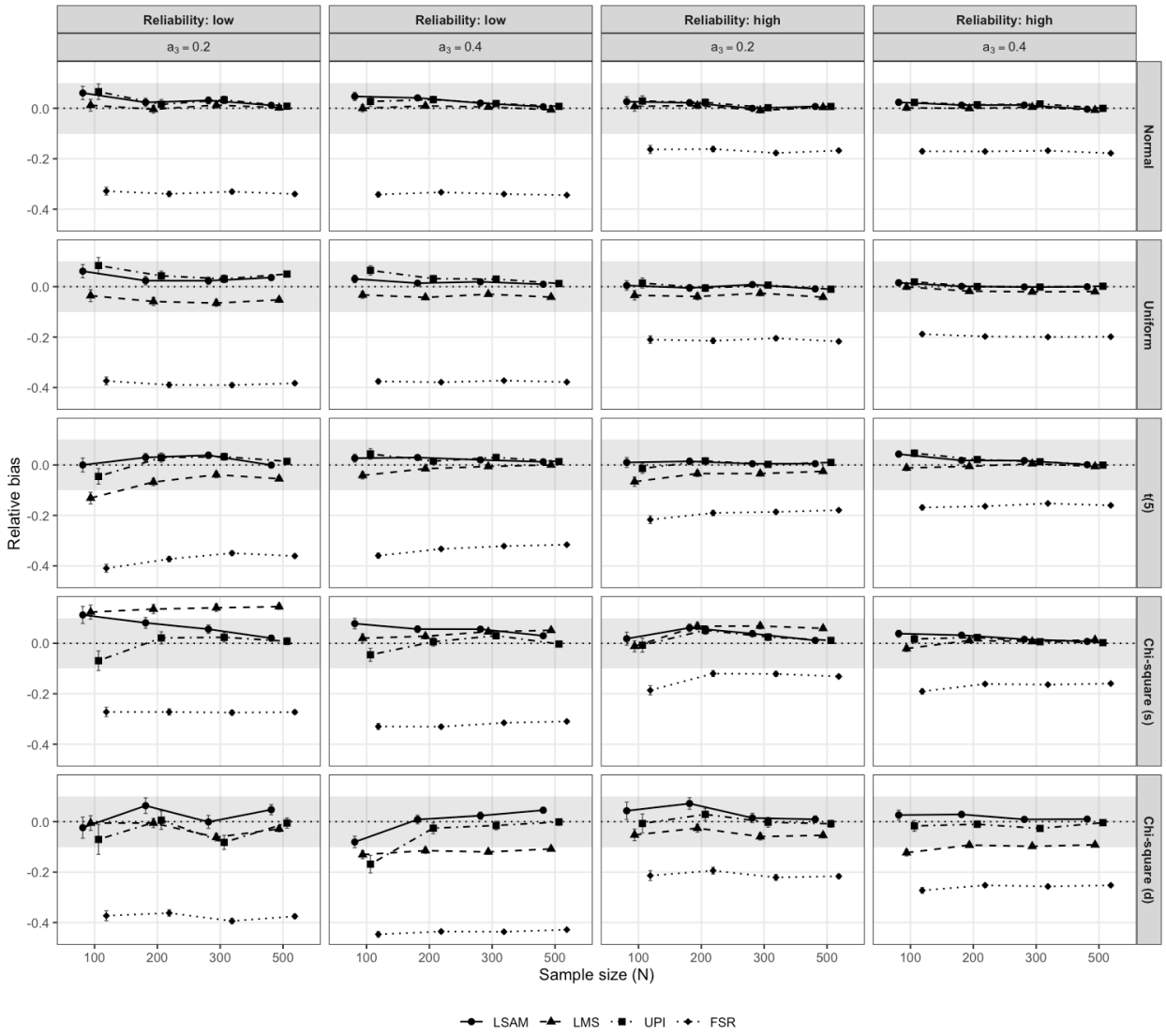
Relative Bias of the imm (a_3b) Under the Correctly Specified Condition Across Sample Sizes, Reliability, Interaction Values, and Distributions



Note. The gray band marks the acceptable region for the measure, and the error bars represent the Monte Carlo standard error (MCSE). imm = index of moderated mediation; LSAM = local structural-after-measurement; LMS = latent moderated structural equations; UPI = unconstrained product indicator.

Figure 3

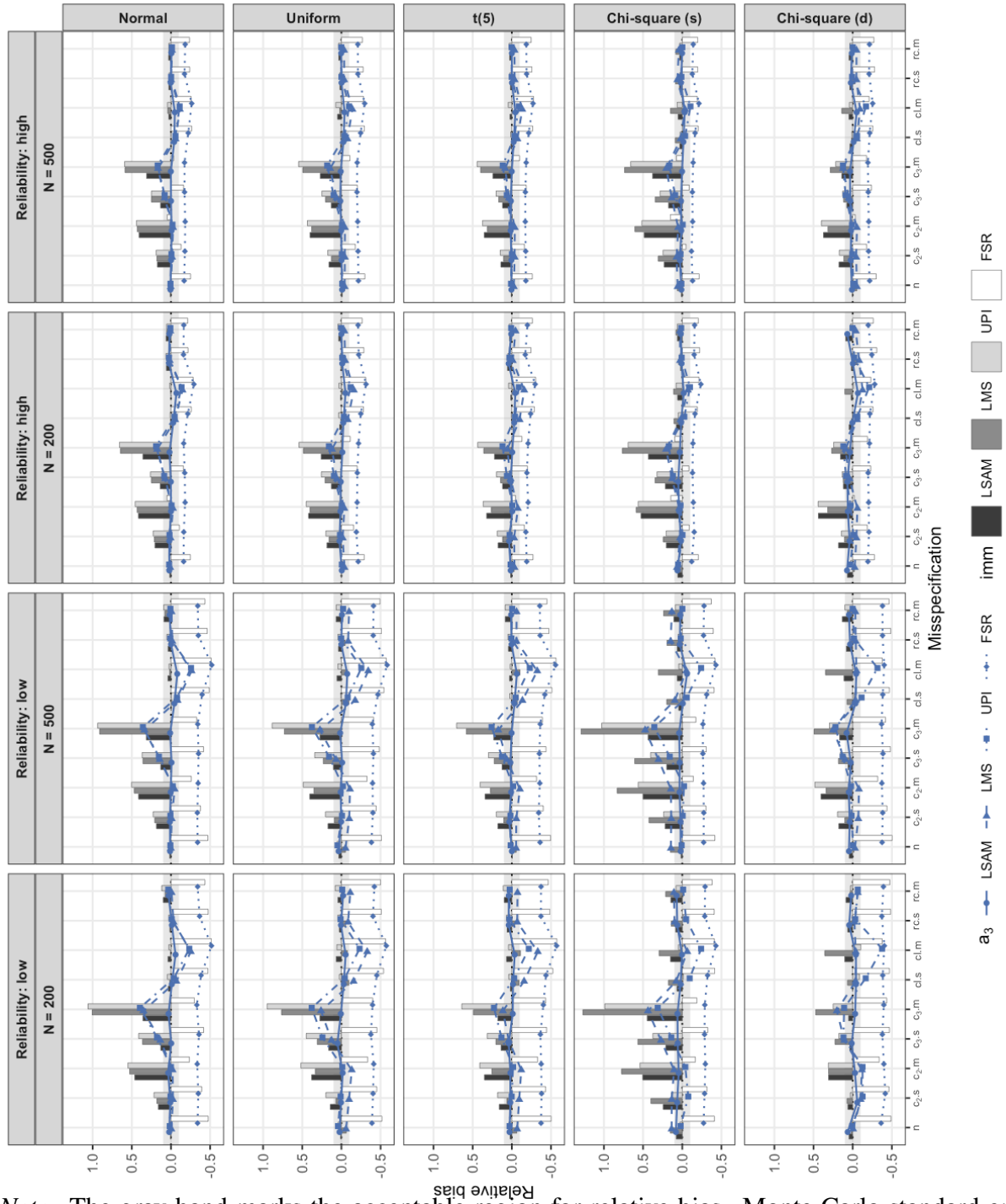
Relative Bias of the Interaction Coefficient (a_3) Under the Correctly Specified Condition Across Sample Sizes, Reliability, Interaction Values, and Distributions



Note. The gray band marks the acceptable region for the measure, and the error bars represent the Monte Carlo standard error (MCSE). LSAM = local structural-after-measurement; LMS = latent moderated structural equations; UPI = unconstrained product indicator.

Figure 4

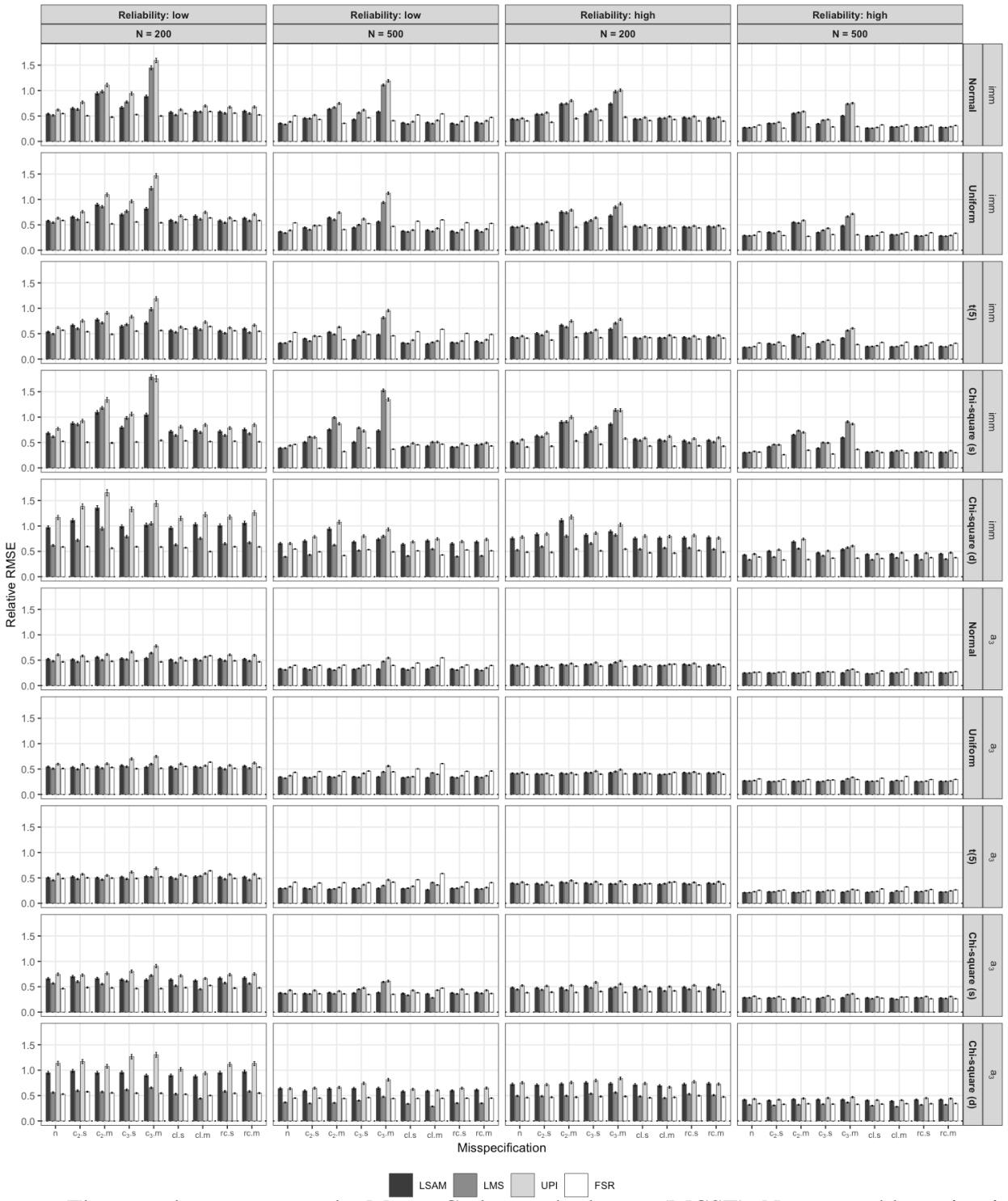
Relative Bias of the imm (a_3b) and the Interaction Coefficient (a_3) Across Distributions, Reliability, and (Mis)specifications for N = 200 and 500 at a_3 = 0.2



Note. The gray band marks the acceptable region for relative bias. Monte Carlo standard errors (MCSEs) are not shown. imm = index of moderated mediation; LSAM = local structural-after-measurement; LMS = latent moderated structural equations; UPI = unconstrained product indicator; n = no misspecification.

Figure 5

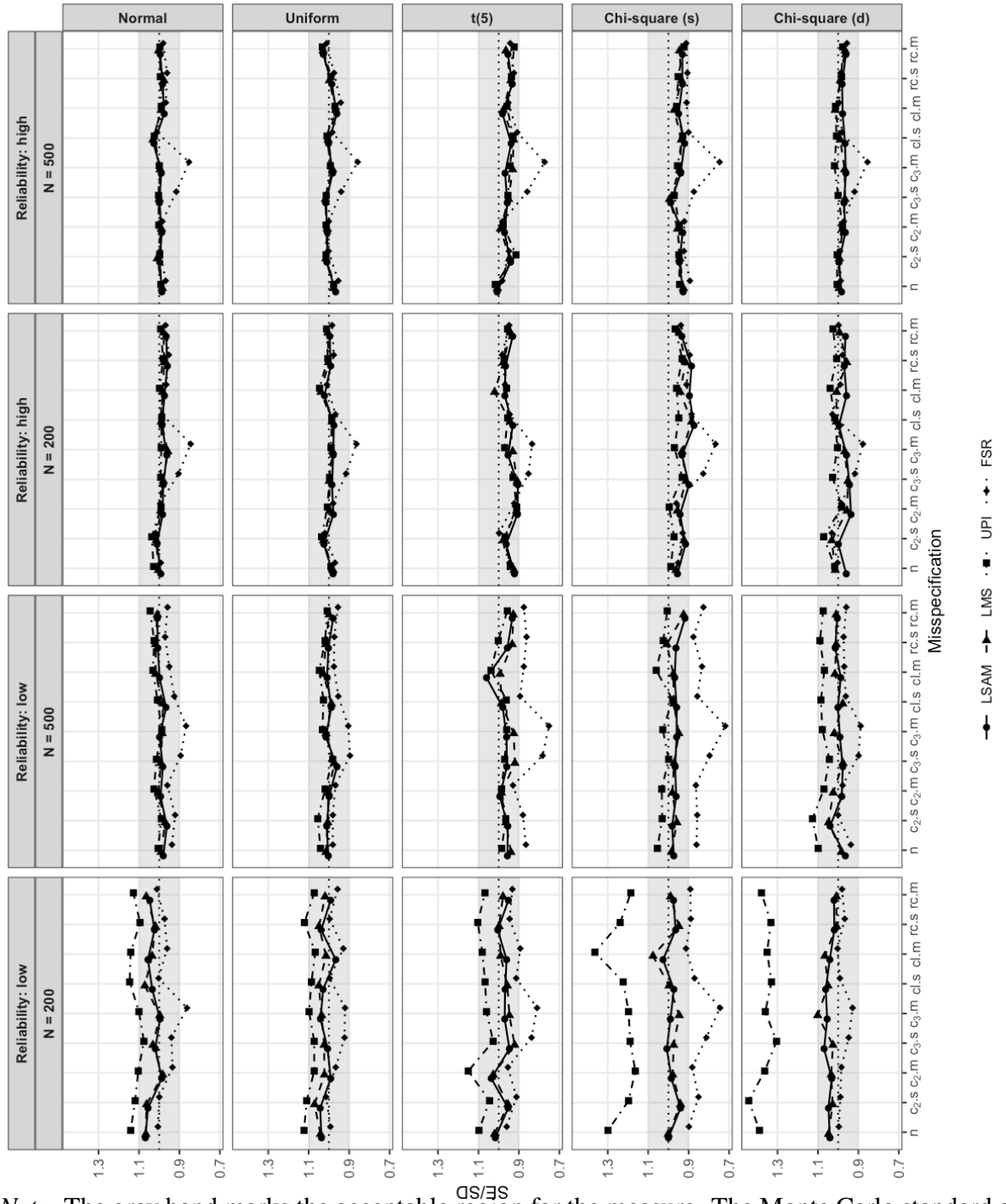
Relative RMSE of the imm (a_3b) and the Interaction Coefficient (a_3) Across Distributions, Reliability, and (Mis)specifications for $N = 200$ and 500 at $a_3 = 0.2$



Note. The error bars represent the Monte Carlo standard error (MCSE). No acceptable region is defined for this measure. imm = index of moderated mediation; LSAM = local structural-after-measurement; LMS = latent moderated structural equations; UPI = unconstrained product indicator; n = no misspecification.

Figure 6

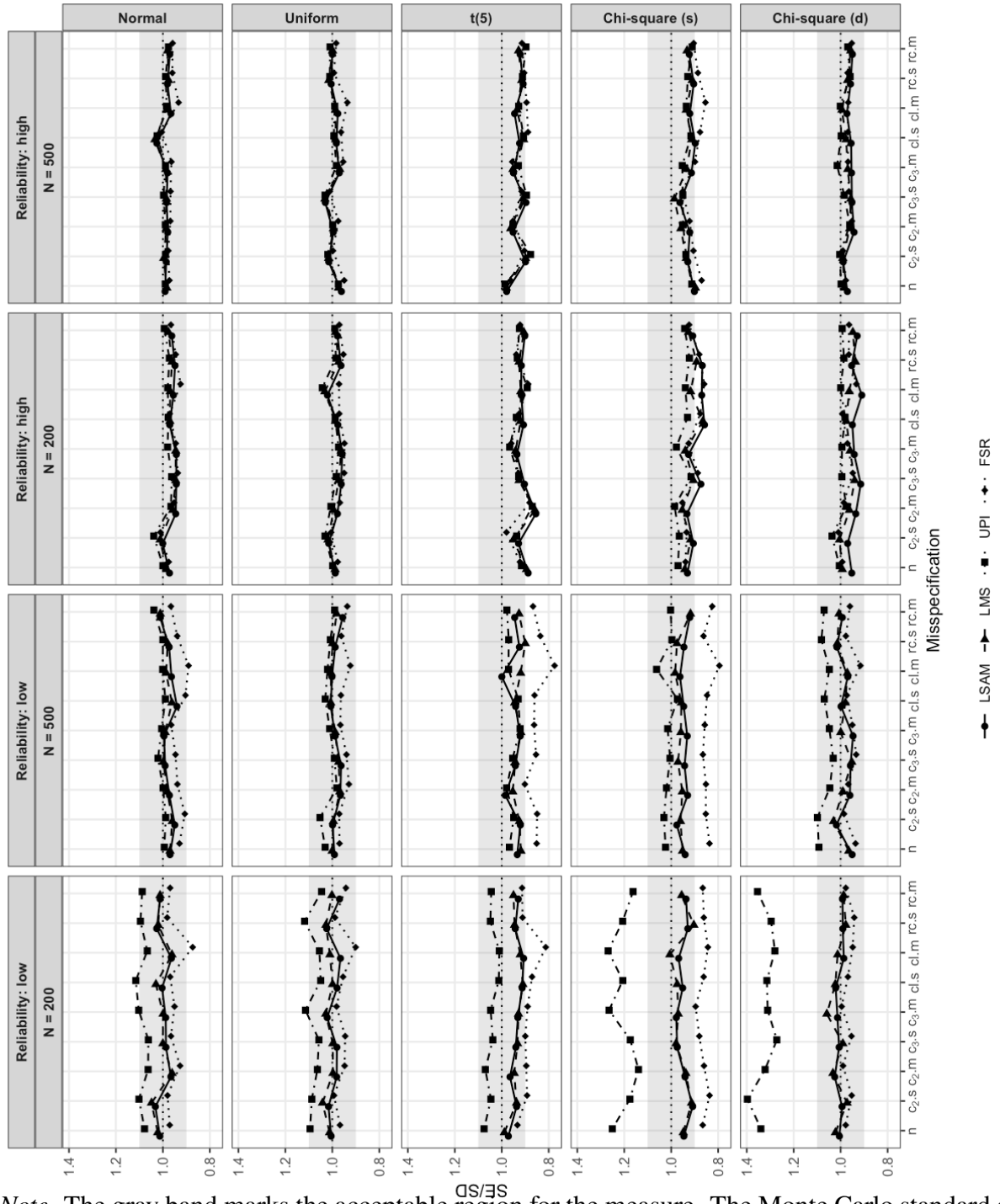
SE/SD of the imm (a_3b) Across Distributions, Reliability, and (Mis)specifications for $N = 200$ and 500 at $a_3 = 0.2$



Note. The gray band marks the acceptable region for the measure. The Monte Carlo standard error (MCSE) is not computed for the SE/SD ratio. imm = index of moderated mediation; LSAM = local structural-after-measurement; LMS = latent moderated structural equations; UPI = unconstrained product indicator; n = no misspecification.

Figure 7

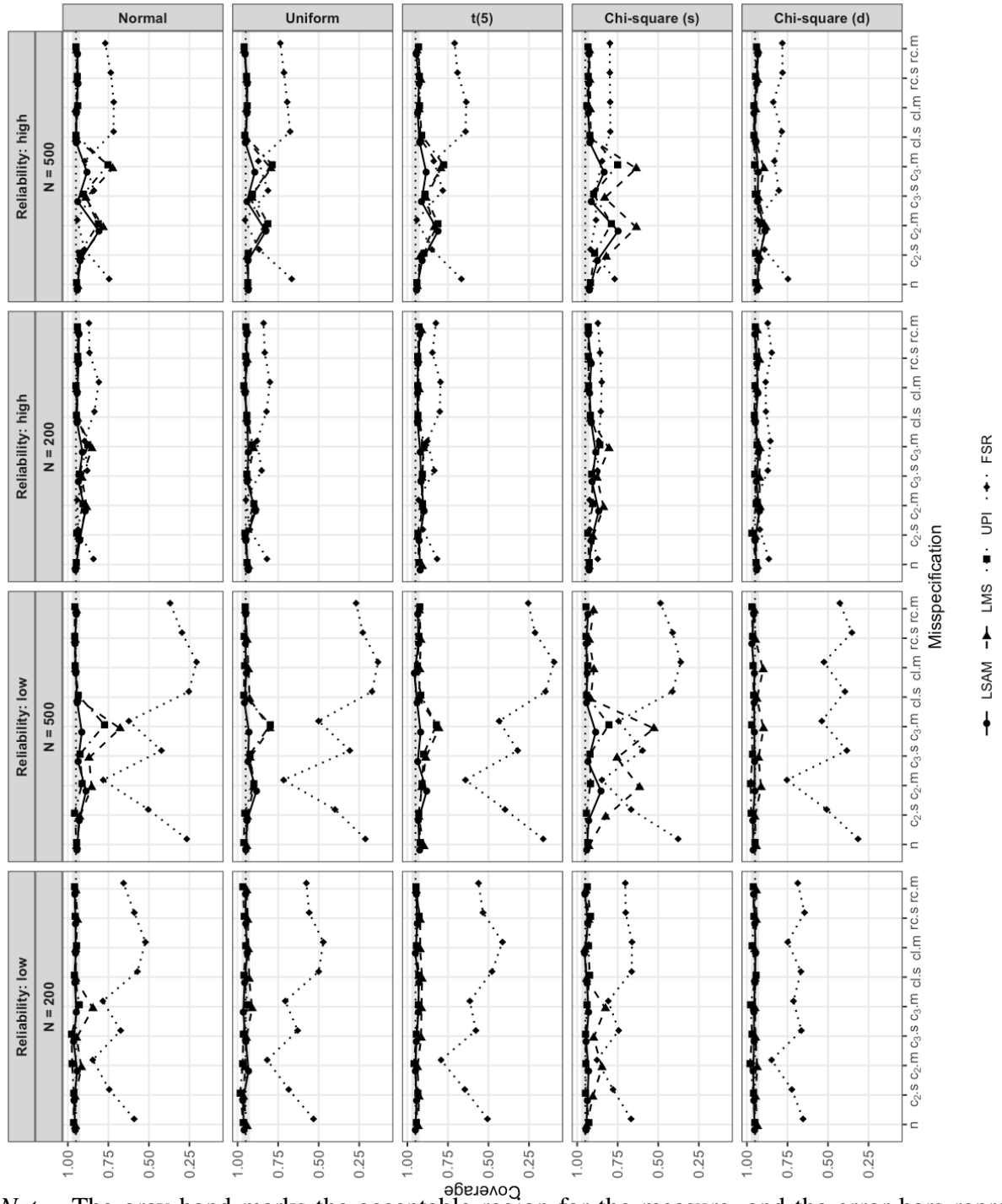
SE/SD of the Interaction Coefficient (a_3) Across Distributions, Reliability, and (Mis)specifications for $N = 200$ and 500 at $a_3 = 0.2$



Note. The gray band marks the acceptable region for the measure. The Monte Carlo standard error (MCSE) is not computed for the SE/SD ratio. LSAM = local structural-after-measurement; LMS = latent moderated structural equations; UPI = unconstrained product indicator; n = no misspecification.

Figure 8

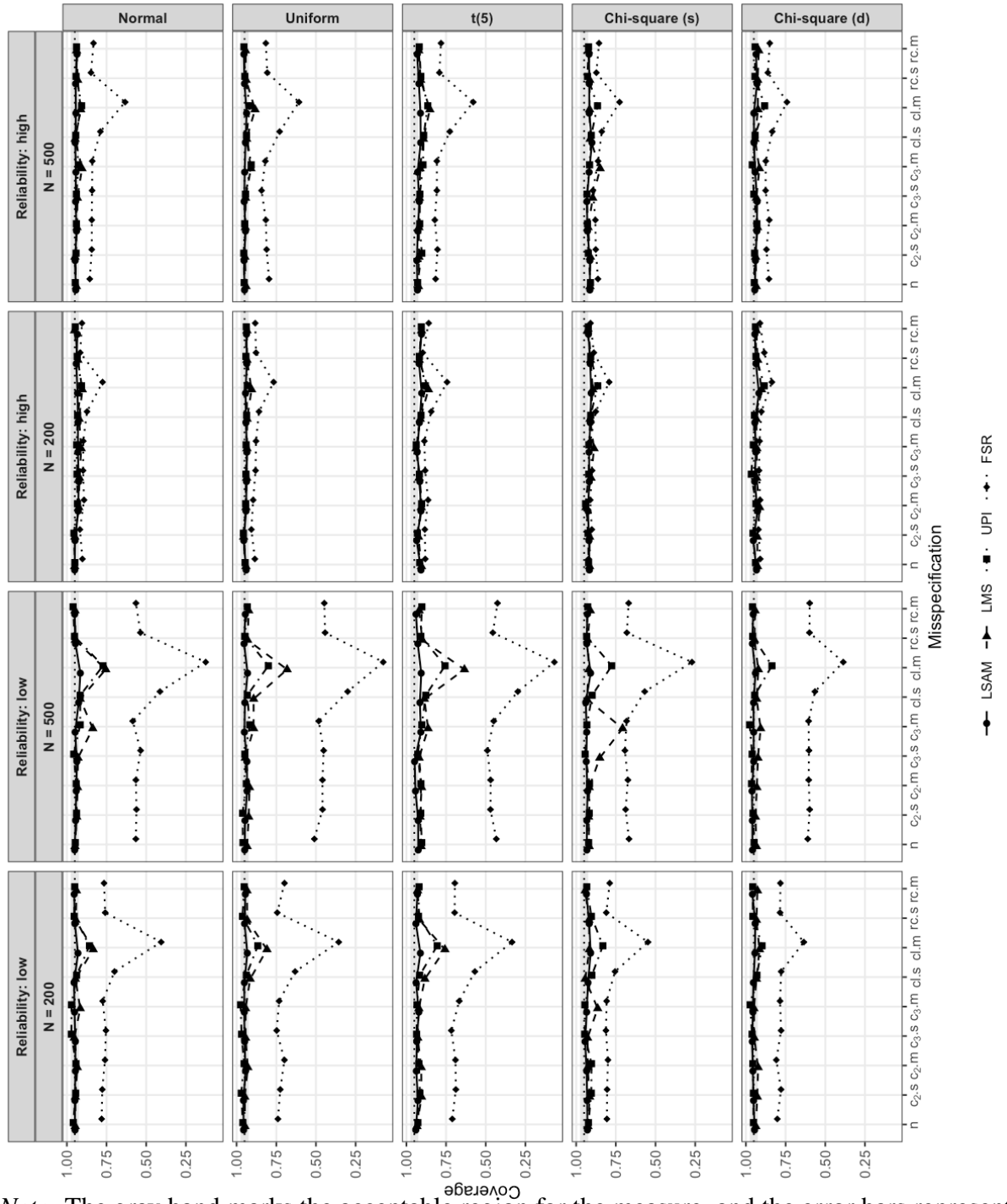
Coverage of the imm (a_3b) Across Distributions, Reliability, and (Mis)specifications for $N = 200$ *and* 500 *at* $a_3 = 0.2$



Note. The gray band marks the acceptable region for the measure, and the error bars represent the Monte Carlo standard error (MCSE). imm = index of moderated mediation; LSAM = local structural-after-measurement; LMS = latent moderated structural equations; UPI = unconstrained product indicator; n = no misspecification.

Figure 9

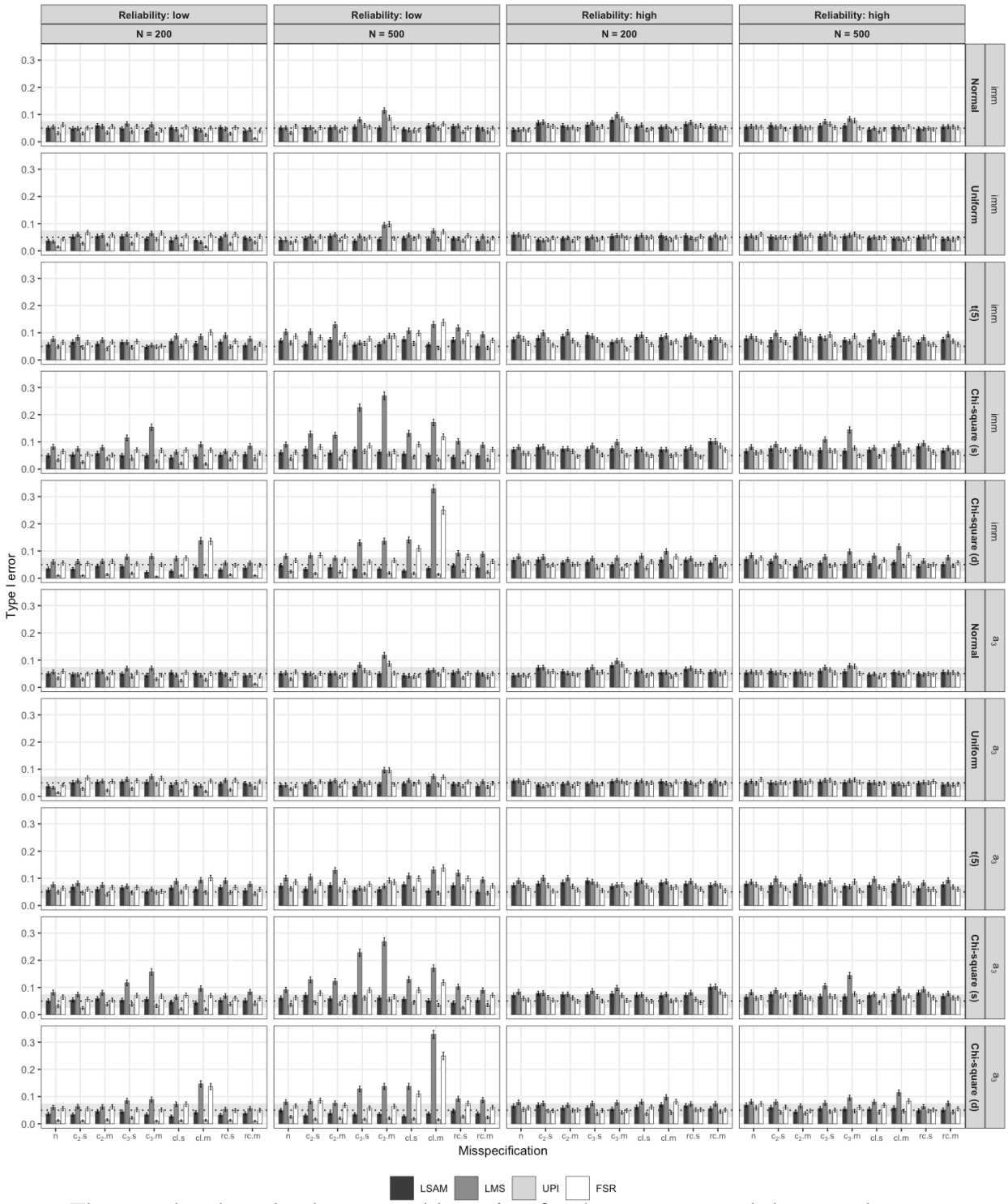
Coverage of the Interaction Coefficient (a_3) Across Distributions, Reliability, and (Mis)specifications for $N = 200$ and 500 at $a_3 = 0.2$



Note. The gray band marks the acceptable region for the measure, and the error bars represent the Monte Carlo standard error (MCSE). LSAM = local structural-after-measurement; LMS = latent moderated structural equations; UPI = unconstrained product indicator; n = no misspecification.

Figure 10

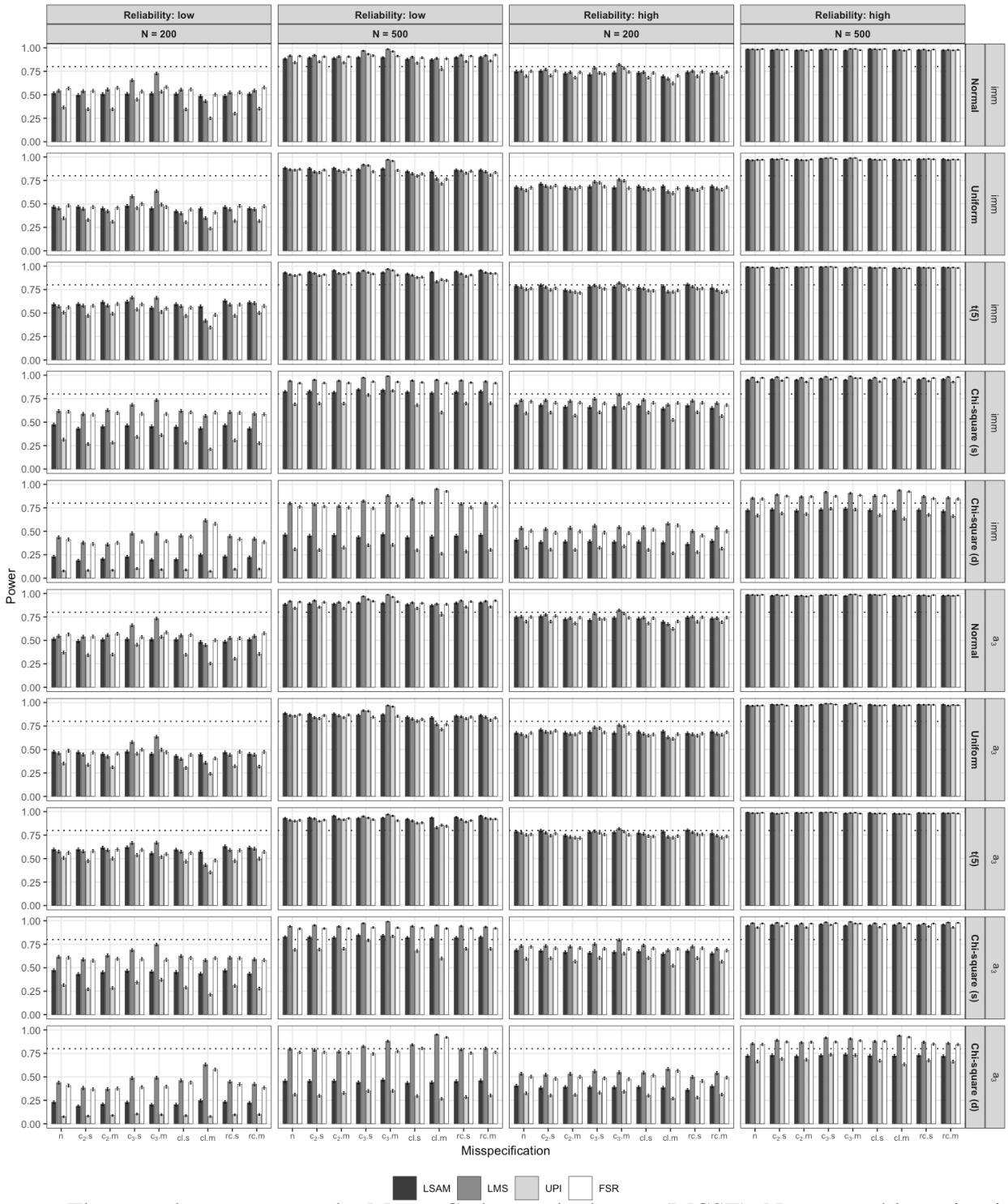
Type I Error of the imm (a_3b) and the Interaction Coefficient (a_3) Across Distributions, Reliability, and (Mis)specifications for $N = 200$ and 500 at $a_3 = 0$



Note. The gray band marks the acceptable region for the measure, and the error bars represent the Monte Carlo standard error (MCSE). imm = index of moderated mediation; LSAM = local structural-after-measurement; LMS = latent moderated structural equations; UPI = unconstrained product indicator; n = no misspecification.

Figure 11

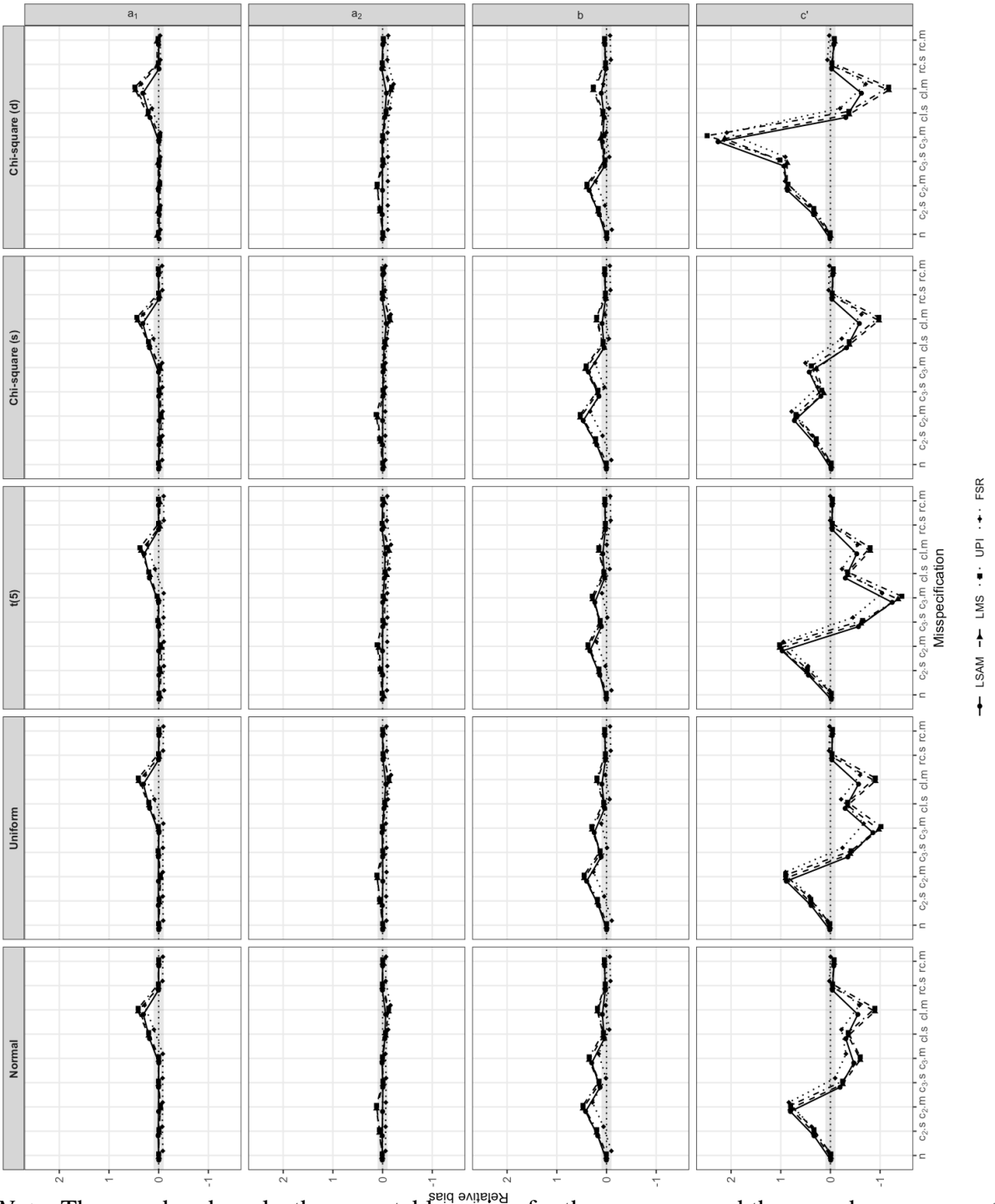
Power for the imm (a_3b) and the Interaction Coefficient (a_3) Across Distributions, Reliability, and (Mis)specifications for $N = 200$ and 500 at $a_3 = 0.2$



Note. The error bars represent the Monte Carlo standard error (MCSE). No acceptable region is defined for this measure. imm = index of moderated mediation; LSAM = local structural-after-measurement; LMS = latent moderated structural equations; UPI = unconstrained product indicator; n = no misspecification.

Figure 12

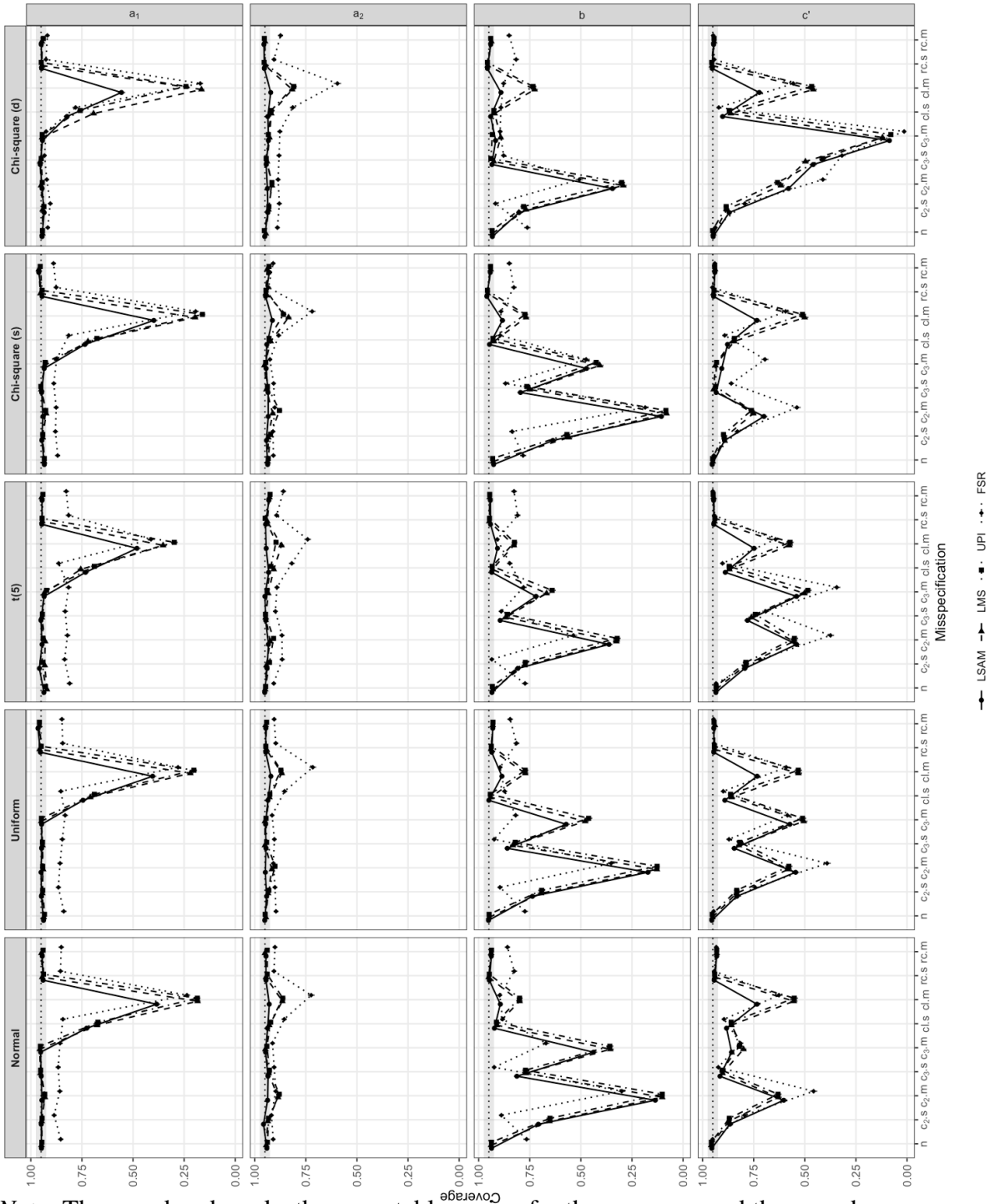
Relative Bias of the Structural Coefficients (a_1, a_2, b, c') Across Distributions and (Mis)specifications at High Reliability, $N = 500$, and $a_3 = 0.2$



Note. The gray band marks the acceptable region for the measure, and the error bars represent the Monte Carlo standard error (MCSE). LSAM = local structural-after-measurement; LMS = latent moderated structural equations; UPI = unconstrained product indicator; n = no misspecification.

Figure 13

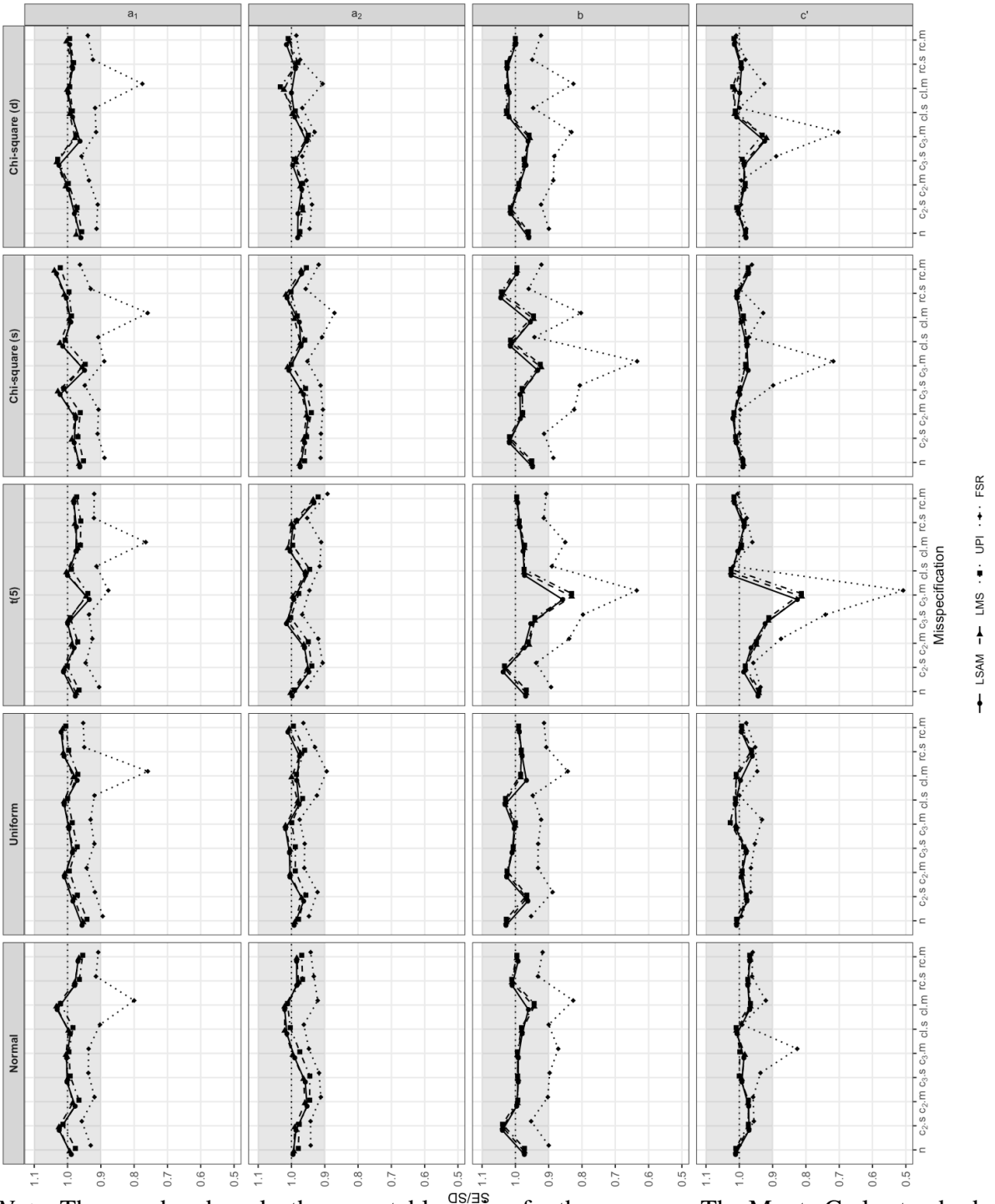
Coverage of the Structural Coefficients (a_1, a_2, b, c') Across Distributions and (Mis)specifications at High Reliability, $N = 500$, and $a_3 = 0.2$



Note. The gray band marks the acceptable region for the measure, and the error bars represent the Monte Carlo standard error (MCSE). LSAM = local structural-after-measurement; LMS = latent moderated structural equations; UPI = unconstrained product indicator; n = no misspecification.

Figure 14

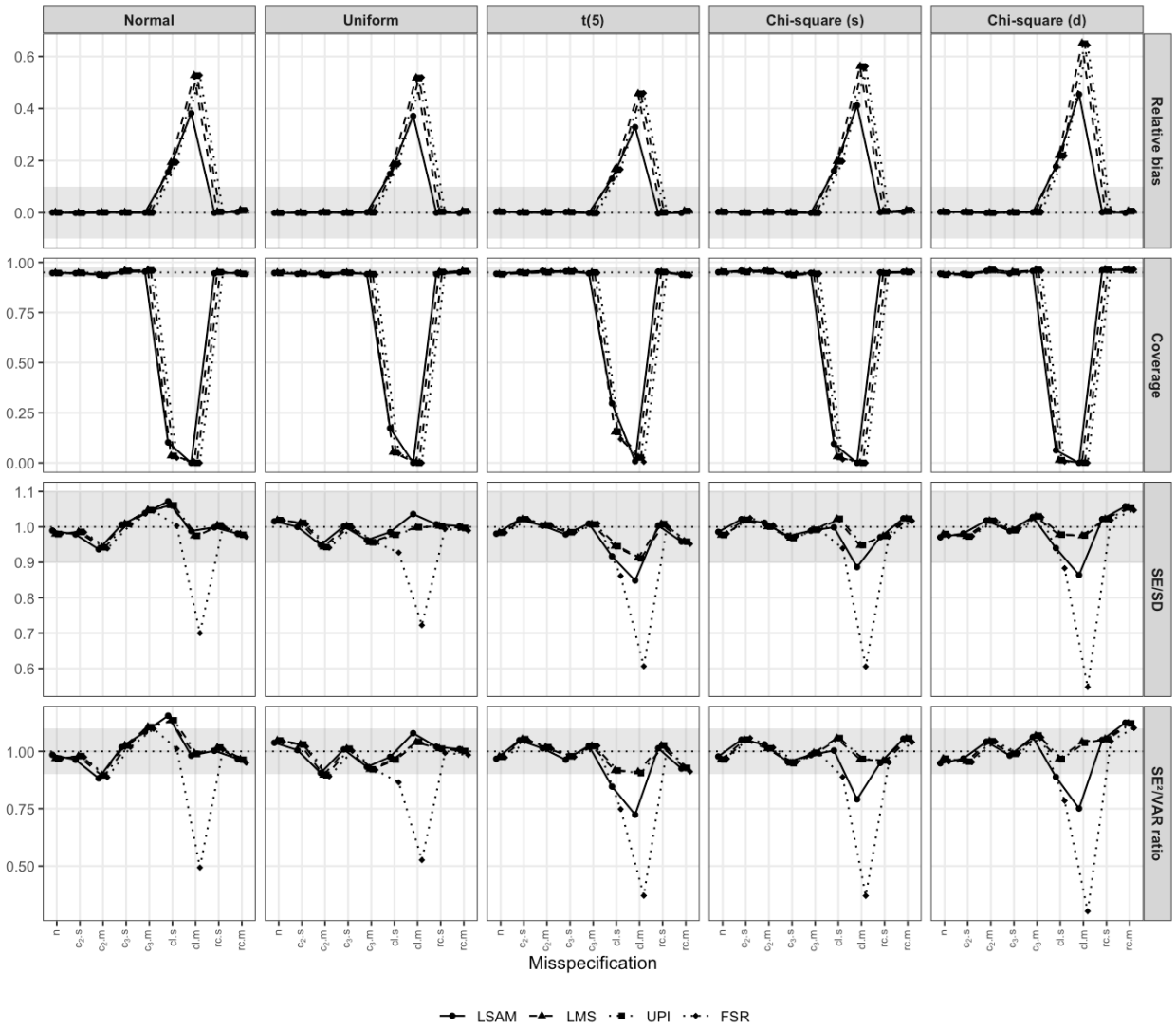
SE/SD of the Structural Coefficients (a_1, a_2, b, c') Across Distributions and (Mis)specifications at High Reliability, $N = 500$, and $a_3 = 0.2$



Note. The gray band marks the acceptable region for the measure. The Monte Carlo standard error (MCSE) is not computed for the SE/SD ratio. LSAM = local structural-after-measurement; LMS = latent moderated structural equations; UPI = unconstrained product indicator; n = no misspecification.

Figure 15

Performance of the Cross-Loaded Mediator Indicator Loading (m_3) Across All Measures, Distributions, and (Mis)specifications at High Reliability, $N = 500$, and $a_3 = 0.2$



Note. The gray band in each panel marks the acceptable region for that measure. LSAM = local structural-after-measurement; LMS = latent moderated structural equations; UPI = unconstrained product indicator; n = no misspecification.

Appendix

Calibration and Data Generation

This appendix gives the formulas behind the data-generating mechanism. The calibration that fixes the variance parameters comes first, followed by the three-stage data generation that uses them.

Calibration

The structural disturbance variances and indicator error variances were fixed by the following closed-form calibration, computed once under the normal baseline.

Structural residual variances

The disturbance variances ψ_M, ψ_Y are fixed so that each structural equation explains a fraction $R^2 \approx 0.30$ of its outcome's variance at the central interaction level $a_3 = 0.2$ under the normal baseline. Writing the systematic part of M as $S_M = a_1 X + a_2 W + a_3(XW)$, its variance needs only second moments,

$$\mathrm{Var}(S_M) = a_1^2 + a_2^2 + 2a_1 a_2 \rho + a_3^2(1 + \rho^2),$$

and the disturbance is the remainder that brings the signal share to R^2,

$$\psi_M = \mathrm{Var}(S_M)\,\frac{1 - R^2}{R^2}, \qquad \psi_Y = \mathrm{Var}(bM + c'X)\,\frac{1 - R^2}{R^2},$$

with $\mathrm{Var}(bM + c'X) = b^2\,\mathrm{Var}(M) + c'^2 + 2bc'\,(a_1 + a_2\rho)$. Fixed at these values, the population variances of M and Y are closed-form functions of the parameters,

$$\mathrm{Var}(M) = \mathrm{Var}(S_M) + \psi_M, \qquad \mathrm{Var}(Y) = b^2\,\mathrm{Var}(M) + c'^2 + 2bc'\,(a_1 + a_2\rho) + \psi_Y,$$

evaluated once under the normal baseline and held fixed across distributions, so that R^2 and reliability drift only through the design factors.

Indicator reliabilities

To hit a target reliability $\omega \in \{0.5, 0.7\}$, the error variance of indicator j is set from the latent variance $V = \mathrm{Var}(\eta)$,

$$\theta_{\eta j} = \lambda_j^2 V \frac{1-\omega}{\omega}.$$

For X, W the variance is exactly 1; for M, Y it is the closed form above, fixed at the normal baseline, so the realized reliability drifts mildly across distributions.

Calibrating the misspecification magnitudes

The *small* and *medium* labels are standardized coefficients, $t \in \{0.25, 0.50\}$. For an omitted predictor p added to a baseline outcome y_0, the magnitude is the fully standardized regression coefficient of p,

$$t = \frac{\beta \ \mathrm{sd}(p)}{\mathrm{sd}(y_0 + \beta p)}.$$

Squaring and substituting $\mathrm{Var}(y_0 + \beta p) = \mathrm{Var}(y_0) + \beta^2 \mathrm{Var}(p) + 2\beta \mathrm{Cov}(p, y_0)$ yields a quadratic $A\beta^2 + B\beta + C = 0$ with

$$A = \mathrm{Var}(p)\,(1 - t^2), \quad B = -2t^2 \mathrm{Cov}(p, y_0), \quad C = -t^2 \mathrm{Var}(y_0).$$

Since $A > 0$ and $C < 0$ the roots have opposite sign; the positive omitted path is the positive root,

$$\beta = \frac{t^2 \mathrm{Cov}(p, y_0) + \sqrt{t^4 \mathrm{Cov}(p, y_0)^2 + t^2(1 - t^2)\ \mathrm{Var}(p)\ \mathrm{Var}(y_0)}}{\mathrm{Var}(p)\,(1 - t^2)}.$$

The three moments are estimated from one large ($n = 10^6$) sample per distribution and a_3 level. The predictor p and baseline y_0 differ by misspecification,

$$\begin{aligned} c_2: &\quad p = W, &\quad y_0 &= bM + c'X + \zeta_Y, \\ c_3: &\quad p = XW, &\quad y_0 &= bM + c'X + \zeta_Y, \\ \texttt{cl}: &\quad p = X, &\quad y_0 &= m_3 = \lambda_3 M + \delta_{m_3}. \end{aligned}$$

c_2 and c_3 are reliability independent; `cl` depends on ω because its baseline m_3 carries measurement noise. `rc` never enters the calibration: it is a residual correlation and equals its target in every cell.

Holding Y reliability fixed under c_2 and c_3

Adding an omitted structural path inflates $\mathrm{Var}(Y)$ by $\kappa = \mathrm{Var}(y_0 + \beta p)/\mathrm{Var}(y_0)$, which would raise Y's indicator reliability as a side effect. Scaling the Y-indicator noise by the same κ makes signal and noise move together, so κ cancels and the reliability returns to its no-misspecification value,

$$\omega_j = \frac{\kappa\,\lambda_j^2 V}{\kappa\,\lambda_j^2 V + \kappa\,\theta_j} = \frac{\lambda_j^2 V}{\lambda_j^2 V + \theta_j}.$$

The latent effect size is untouched; only the Y indicators are made correspondingly noisier. The measurement level misspecifications `cl` and `rc` act on the indicators directly and are not compensated, since that contamination is itself the misspecification.

Data generation

Using the calibrated variances above, data were generated in three stages: the correlated exogenous predictors, the dependent latent variables through the structural model, and the observed indicators through the measurement model.

Stage I: Exogenous predictors

The correlated predictors (X, W) were generated with the vine-to-anything (VITA) method (Grønneberg et al., 2022), as implemented in the *covsim* package, with sampling via *rvinecopulib* (Nagler & Vatter, 2025). They were specified to have mean 0, variance 1, and the covariance structure

$$\Phi = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0.3 \\ 0.3 & 1 \end{bmatrix},$$

so that the conditions differ only in distributional shape. Under the normal condition, the predictors had Gaussian marginals with the specified covariance structure. Under the uniform condition, they had symmetric uniform marginals $\mathrm{U}(-\sqrt{3}, \sqrt{3})$, which have mean 0 and variance 1. Under the t_5 condition, they had Student t marginals with five degrees of freedom, rescaled by $\sqrt{5/3}$ to unit variance. The two χ^2 conditions used χ^2_2 marginals, centered by their mean (2) and

scaled by their standard deviation (2) to mean 0 and variance 1. In the same-direction condition both predictors are right-skewed, whereas in the opposite-direction condition W is negated so that X and W are skewed in opposite directions, with the sampled correlation sign flipped to restore $\mathrm{Cor}(X, W) = \rho$. In every condition, VITA constructed a vine copula (Gaussian under the normal condition, Clayton otherwise) to impose the target correlation while preserving the marginals, with the t_5 and χ^2 draws standardized afterwards as described.

Stage II: Structural model

The dependent latent variables were generated from the structural equations

$$M = a_1 X + a_2 W + a_3 (X \times W) + \zeta_M, \qquad Y = bM + c' X + \zeta_Y,$$

with independent Gaussian disturbances $\zeta_M \sim \mathcal{N}(0, \psi_M)$ and $\zeta_Y \sim \mathcal{N}(0, \psi_Y)$. The disturbance variances ψ_M, ψ_Y take the calibrated values so that each equation explains $R^2 \approx 0.30$ of its outcome's variance at the central interaction level under the normal baseline.

Stage III: Measurement model

Each latent $\eta \in \{X, W, M, Y\}$ was measured by four indicators,

$$z_{\eta j} = \lambda_j \, \eta + \delta_{\eta j}, \qquad \lambda = (1, 0.8, 0.7, 0.6),$$

with independent Gaussian errors $\delta_{\eta j} \sim \mathcal{N}(0, \theta_{\eta j})$. The error variances $\theta_{\eta j}$ take the calibrated values that achieve the target reliability $\omega \in \{0.5, 0.7\}$. Only the exogenous predictors varied in distribution. The structural disturbances and measurement errors were Gaussian in every condition.